MultiQC Report Summary – DSMZ Cell Line RNA-seq (FastQC)

Overview

Report Generated: August 2025
Total Samples: 367 paired-end samples (~184 actual cell lines)
Tool Used: FastQC via MultiQC
Read Length: 151 bp (uniform across all samples)
Data Source: DSMZ and associated collections (e.g., breast cancer, retinoblastoma, neuroblastoma)

Key Quality Metrics

Strengths:
- Read Quality: High per-base quality scores for most samples
- Read Length: Uniform 151 bp reads, ideal for standard RNA-seq pipelines
- GC Content: Normal range (~47–52%)
- Read Depth: 6–70 million reads per sample, offering sufficient coverage for classification

Concerns:
- High Duplication Rates (31–76%, most at 50–70%)
- FastQC Module Failures (0–36% per sample)
- Adapter Contamination in a subset of samples

Sample Quality Categories

■ Best: Low duplication (<50%), few FastQC failures
■ Moderate Concerns: Duplication (55–65%), some failures
■ Needs Attention: High duplication (≥70%), adapter contamination

Examples:
- Best: NALM_6, HPB_ALL, NG-13640 samples with <50% duplicates
- High Duplication: NG-13640_KASUMI_1_RNA (76.7%), K_562_RNA (73.3%), NG-A2798_GB2 (72.9%)

Biological Context

Cancer Types Represented:
- Leukemia/Lymphoma: HL60, NALM, JURKAT, K-562
- Breast Cancer: MCF-7, MDA-MB-231, T-47D
- Retinoblastoma: RBL series, WERI-RB1, Y-79
- Neuroblastoma: IMR32, KELLY, LAN series

Significance:
- Enhances multi-class classification
- Provides robust training data
- Introduces batch effect and class imbalance challenges

Recommendations

Immediate Actions:

- Investigate high-duplicate samples
- Trim adapters
- Consider deduplication

For Classification:
- Use QC covariates (duplication %, depth, batch)
- Filter outliers if needed

Classification Strategy

Strengths:
- Multi-class potential
- Well-characterized cell lines
- Strong ML training data

Challenges:
- Batch effects
- Imbalanced class sizes
- Distinguishing biological vs. technical variance

Next Steps

1. Proceed with classification analysis
2. Trim adapters and clean up outliers
3. Include QC metrics in modeling
4. Explore clustering or unsupervised methods

Bottom Line

You have a high-quality, biologically rich dataset for classification.
FastQC confirms this data is ready for robust, multi-class machine learning analysis.